

ESD-CONTROL IONISERS For CLASS-0 DEVICES



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Electrostatic discharge (ESD) protection is the primary failure-control mechanism implemented by organisations handling ESD-sensitive components (ESDS). Various industry statistics point to 28 per cent of electronic component failures to ESD. However, with increasing use of Class-0 ESDs in electronics assemblies, this failure rate is expected to increase significantly unless adequate failure-control mechanisms are implemented.

Ionisers

While several techniques like continuous monitoring and additional redundant grounding are deployed for Class-0 components, ionisers are the most effective solution for protection while handling Class-0 components.

Ionisers are used for neutralising any undesired electric charge present on ESDS or mandatory insulator present in an ESD-protected area (EPA). These come in various forms that can be simple desktop fan type, horizontally-mounted fan on table-top or advanced ionisers with feedback mechanism and provision for continuous performance through connection to a computer or some other data-logging system.

An ioniser fan is directed to the area where work is being performed to ensure required charge neutralisation. Also, for feedback-controlled ionisers, feedback sensors are spaced as per the manufacturer's recommendations on the work surface.

Basic working principle

While some ioniser equipment have many advanced features, the basic working principle of the equipment is detailed in Fig. 1.

A typical ioniser has transformer (part 1 in Fig. 1) connected to power supply, which amplifies charges and feeds

to a metal mesh (part 3), which gets charged. A fan (part 2) forces charged ions (part 4) to be subjected on EPA (part 5), which can be a work-station or a larger EPA.

However, DC-type ionisers use separate sources for generation of positive and negative voltages.

Key features to be considered

Some key criterion when selecting an ESD ioniser are:

Identify class of ESDS component being handled. Depending on the class of component being selected, ionising capacity is defined. Else, this can say whether an ESD control ioniser is required at all.

Activity being performed. This includes points including a manual or an automated operation and different material of the surfaces coming in contact (for example, sponge surface coming in contact with plastic-body component). Sufficient samples need to be considered of peak voltage generation due to all activities.

Sensitivity of performance. Cycling nature of charge generation on the work-surface will determine precision of the feedback mechanism of ionisers.

Area to be covered. Spread of the work-surface to be covered determines whether the equipment should be desktop fan-type or curtain-type ceiling-mounted.

Specialised work or re-work. Some of the work requires focussed ionisation depending on the type of work being done in a localised area, which may involve some static generating materials, too.

Power back-up. Good ionisers have automatic shutdown protection inbuilt within the equipment.

Charge balance indicator. Ionisers also have a charge-balance indicator with provision for connection to an alarm and a data-logger for continuous monitoring.

Modularisation of key elements of the ioniser. The equipment should be modular and easy to dismantle and assemble since ionisers require regular maintenance.

Fig. 1: Basic working principle of ESD control ioniser

